Muhammad Abdullah

Python backend and AI engineer

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Available for freelance and part-time work now, and for full-time roles from June 2027, on-site or remote.

Education

BS Computer Engineering · COMSATS University Islamabad, Lahore Campus — Sep 2023 – Jun 2027
Coursework: VLSI Design, Digital System Design, Data Communication and Networks, Artificial Intelligence, Machine Learning.
Final-year project: machine learning that detects RF jamming in 2.4 GHz wireless networks.

Intermediate (ICS), Grade A · Punjab Group of Colleges — 2021 – 2023

Projects

Bank Reconciliation API `FastAPI · Pandas · PostgreSQL · pytest` — 2026
Matches a bank statement to a ledger. Given two CSV files, a background job matches them, allowing two days between dates, and returns the matched and unmatched transactions and the balance difference in about 2 seconds. Code

AI Sales Copilot `FastAPI · Gemini · Supabase · SQLAlchemy` — 2026
A sales assistant for marketing agencies. Gemini scores every incoming lead from 0 to 100 and suggests how to approach it, and each lead has its own AI chat that drafts follow-up emails or proposals. Live app · Code

AEGIS-SWARM Threat Triage · team project `scikit-learn · OpenCV · Streamlit · Supabase` — 2026
Catches phishing in SMS, emails, URLs and QR codes. NLP agent (TF-IDF plus 13 security heuristics, soft-voting ensemble, 24,206 training messages): 94.34% test accuracy. QR agent (OpenCV decoding, URL classifier): 96.44%. Demo · Code · Models

FlumenIntel `Gradio · Groq · Gemini · Plotly` — 2025
A river health dashboard. It turns field readings into water-quality, habitat and erosion scores, adds a brightness-based turbidity score from a satellite image, and has an LLM write a report that downloads as a PDF. Code

Sparsity-Aware Processing Element · team of three `VHDL · Quartus · Microwind` — 2026
Power-gated processing element in VHDL (8-bit multiply, 32-bit accumulate), the basic block of an 8×8 systolic array for neural-network inference. On a zero weight a sleep transistor powers the multiplier down. Code

Smaller projects

- Data Normalizer API (2026): async service that cleans messy CSV, Excel and JSON files and returns an audit report.
- FastAPI JWT Auth & CRUD API (2026): JWT tokens, OAuth2 password flow, bcrypt hashing and owner-only edit and delete.
- Fake News Detector (2025, hackathon): scores an article's fake-news likelihood and sentiment, and explains it with an LLM.

Experience

Intern Web Developer · MKY Global Forwarding, remote — Mar 2025 – Jul 2025
Deployed a logistics website for a startup and worked with its developers on other projects in HTML, CSS, JavaScript and SQL.

Microsoft Learn Student Ambassador · COMSATS University, Lahore — Feb 2025 – Jun 2025
Led technical workshops on AI and cloud for students, and helped plan the ambassadors' activities on campus.

C++ Developer Intern · Rhombix Technologies, remote — Jan 2025 – Mar 2025
Built C++ programs around data structures, algorithms and OOP, including a to-do list manager and a Caesar-cipher tool.

Certificates

- **Generative AI Application Developer**, Pakistan Engineering Council with PakAngels (2025): top performer in Cohort 3.
- **AWS AI & ML Scholars: 2026 Challenge Completion**, Udacity and AWS (2026): pre-trained dog-breed classifier project.
- **AWS Cloud Quest: Generative AI Practitioner** and **AWS SimuLearn: Serverless Developer**, AWS (2026), both on Credly.
- Online courses (2025–2026): Generative AI for Everyone (DeepLearning.AI), Create Generative AI Apps on Google Cloud, Programming for Everybody and Python Data Structures (University of Michigan), and 10 DataCamp courses in SQL, Python, APIs, data engineering and cloud computing.

Skills

Languages	Python, SQL, C++, Java, JavaScript / TypeScript
Backend	FastAPI, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, Supabase, Pandas, JWT / OAuth2, pytest
AI and data	Gemini and Groq APIs, RAG, Hugging Face, scikit-learn, TensorFlow / Keras, OpenCV, Gradio, Streamlit
Cloud and DevOps	AWS (Lambda, API Gateway, S3), Docker, Vercel, Render, Git / GitHub
Hardware and RF	VHDL, Verilog, Xilinx Vivado, Quartus, Microwind, GNU Radio, RTL-SDR

Volunteering

IEEE Student Branch · COMSATS Lahore, member. Certificate of Appreciation, Spring 2025. — Feb 2025 – Sep 2025

Youthe · COMSATS Lahore, member. Student-led development events and community service. — Feb 2025 – present